BLOOD BANK MANAGEMENT SYSTEM

SYSTEM VALIDATION REPORT

BBMS Version 1.0 | Evidence-based assessment – Version 3.1

Prepared by	Pallavi Rajkumar Bubanale
Team / contribution	Team 9	System validation evidence contributor
Assessment date	12 November 2026
Evidence package	current.zip – 75 screenshots, plus individually shared screenshots and terminal output
Overall status	PARTIAL SYSTEM VALIDATION – STP EXIT CRITERIA NOT DEMONSTRATED

Document control

Version	Date	Description
2.0	12-Nov-2026	Previous evidence-based report; baseline for this revision.
3.1	12-Nov-2026	Corrected test-case evidence mapping and search-case descriptions; clarified individually shared validation evidence.

1. Executive summary

This report presents an evidence-based system validation assessment of the Blood Bank Management System (BBMS) Version 1.0. It consolidates the earlier 67-screenshot evidence set, the latest archive current(1).zip containing 75 image files, additional screenshots shared individually in the validation conversation, and the terminal output confirming Django password hashing.

The Software Test Plan (STP) identifies 28 system test cases. Based on the evidence available at the assessment date, 16 cases are classified PASS, 2 PARTIAL and 10 NOT RUN. These are evidence-status counts, not a formally measured pass rate. The STP exit criteria cannot be declared satisfied because several planned performance, reliability, portability and security activities were not executed or were not evidenced.

Additional evidence received for this revision demonstrates invalid blood-request form validation, accepted expired-stock exclusion, near-expiry donation display, unauthenticated report-route protection, and password hashing. The rejected-request issuance attempt remains unresolved: the issuance URL displayed the rejected request detail page, so blocking of issuance was not demonstrated.

2. Scope and evidence basis

Validation scope follows the BBMS STP categories: donor management, donation recording, inventory, expiry monitoring, search, blood requests, issuance/fulfilment, reports, performance, reliability, usability, portability and security. The report evaluates the application behaviour shown by the supplied evidence; it does not certify clinical suitability, blood quality, laboratory screening, physical storage equipment, or external hospital integrations.

Evidence sources used:

- Latest uploaded archive current(1).zip (75 screenshot files).
- Earlier uploaded archive current.zip (67 screenshot files), used as the baseline evidence set in v2.0.
- Additional individual screenshots shared in this conversation, including donor required-field validation and invalid-phone validation.
- Terminal output shared in this conversation showing the stored password hash for the synthetic hospital_test account and Is hashed: True.
- BBMS STP and SRS references and the prior v2.0 report as the assessment baseline.

3. Test environment and limitations

Operating system	Windows 11 Home Single Language 25H2 (previously recorded; not re-probed for this revision).
Runtime	Python 3.12.10 and Django 5.2.18 (previously recorded).
Browser	Chrome 157 visible in the screenshots; no second browser was evidenced.
Application	Local development server at http://127.0.0.1:8000/.
Database	Local SQLite used for the latest shell/database checks. Earlier v1.4/v2.0 material referred to a separate PostgreSQL context; do not combine the two runs.
Test data	Synthetic donor, hospital, patient, donation and request records; test

	accounts include hospital_test and bbms_admin.
Source changes	No application source-code changes were made as part of the validation activities described here. Test accounts, groups and records were created locally.
Limitations	No fresh full automated test/coverage run, measured load test, controlled concurrency test, cross-browser run, HTTPS deployment test, SQL injection test, XSS test or formal usability study is evidenced.

4. Result definitions and method

- PASS: the available evidence demonstrates the test objective and expected behaviour.
- PARTIAL: evidence exists, but it does not prove the full objective or leaves an important condition unresolved.
- FAIL: evidence demonstrates an actual mismatch against the expected result. No case is classified FAIL in this revision because no complete, attributable failure record was established for the matrix below.
- NOT RUN: the required test was not performed or no sufficient execution evidence was supplied.

A screenshot of a form is not treated as proof that a transaction succeeded. Where possible, success messages, inventory changes, request status, database verification and follow-on views were considered together. Historical automated test and coverage figures are identified as historical and are not counted as fresh execution.

5. Execution summary

Status	Count	Interpretation
PASS	16	Objective demonstrated by available evidence
PARTIAL	2	Some evidence; full objective not established
FAIL	0	No evidence-classified failure in this matrix
NOT RUN	10	Not executed or insufficient evidence

No formal pass percentage is reported. Partial and not-run cases remain visible and are not counted as passed.

6. STP traceability matrix – 28 system test cases

Test ID	Test objective	Status	Evidence / assessment
TC-DON-01	Donor registration with valid details	PASS	SYS-07, SYS-08
TC-DON-02	Donor registration input validation (required field / invalid phone)	PASS	Individually shared donor-form screenshots — required full-name field validation and invalid phone-number validation messages shown.
TC-DON-03	Donor search and donor update	PASS	SYS-09, SYS-11, SYS-12
TC-DNT-01	Accepted, pending and rejected donation recording / stock effect	PASS	SYS-13–SYS-22
TC-DNT-02	Donation expiry-date validation	PASS	Individually shared donation-form screenshot — browser validation rejected an expiry date earlier than the minimum allowed date (05-Nov-2026).
TC-INV-01	Available inventory calculation and accepted-stock update	PASS	SYS-15, SYS-16, SYS-18, SYS-21, SYS-36
TC-MON-01	Expired donation visibility and exclusion from available stock	PASS	SYS-NEW-02_Expired_Donation_Registration; SYS-NEW-03
TC-MON-02	Near-expiry donation visibility (next 7 days)	PASS	SYS-NEW-06_Near_Expiry_Stock_Verification

TC-SRCH-01	Search for an available blood group and display available units	PASS	SYS-44–SYS-45 — O+ availability search displayed 4 available units in the recorded test state.
TC-SRCH-02	Search for an unavailable blood group and display zero available units	PASS	SYS-46 — A+ availability search displayed 0 units and an unavailable status.
TC-REQ-01	Blood request submission and workflow	PASS	SYS-24–SYS-31, SYS-37
TC-REQ-02	Invalid / missing mandatory blood request fields	PASS	SYS-NEW-01
TC-ISS-01	Successful blood issue, allocation and stock deduction	PASS	SYS-33–SYS-37
TC-ISS-02	Issuance protection for rejected / invalid request	PARTIAL	Rejected request details and issuance URL attempt shown in conversation; issuance URL displayed request details; block not proven
TC-ISS-03	Failed issuance transaction rollback	NOT RUN	No controlled failure / rollback evidence
TC-RPT-01	Reports, history and filters	PASS	SYS-48–SYS-58; SYS-68
TC-PERF-01	Response-time measurement	NOT RUN	No repeatable timing measurements supplied
TC-PERF-02	Concurrent request / issuance behaviour	NOT RUN	No controlled concurrent execution evidence
TC-PERF-03	Load / stress performance	NOT RUN	No load-test results supplied
TC-REL-01	Reliability / recovery behaviour	NOT RUN	No repeatable recovery or fault-injection evidence
TC-USE-01	Usability evaluation	NOT RUN	No structured user observation / usability metrics
TC-PORT-01	Cross-browser / platform compatibility	NOT RUN	Only Chrome evidence; no comparative browser run
TC-SEC-01	Unauthenticated access control	PASS	SYS-NEW-04
TC-SEC-02	Role-based access control	PARTIAL	SYS-59–SYS-69; selected role routes only; group/profile inconsistency observed
TC-SEC-03	Injection / input attack resistance (e.g. SQL injection)	NOT RUN	No controlled injection test evidence
TC-SEC-04	Password storage protection	PASS	SYS-NEW-05; Django PBKDF2-SHA256 hash confirmed
TC-SEC-05	HTTPS / transport security	NOT RUN	Local HTTP environment; no HTTPS deployment evidence
TC-SEC-06	XSS / remaining security verification	NOT RUN	No controlled XSS/security scan evidence

Note: The classification maps the evidence to the closest STP case. Where the exact internal sub-condition cannot be established from the screenshot alone, the assessment remains PARTIAL or NOT RUN rather than being inferred.

7. Key validated workflows and observations

Donor management: Donor registration, donor lookup, donor update and donor history were demonstrated. Individually shared screenshots also show required-field and invalid-phone validation messages; these are cited separately rather than incorrectly mapped to SYS-04 or SYS-05 in the archive.

Donation recording: Accepted, pending and rejected donation records were created and their effect on available stock was checked. A separately shared donation-form screenshot shows browser validation rejecting an expiry date earlier than the permitted minimum.

Inventory and expiry: The latest expired-stock evidence shows accepted donation RIFCYOY541 (O+, 2 units, expiry 01-Nov-2026) under Expired Donations while O+ available stock remained 1. The latest near-expiry evidence shows donation 8NHH0RRBC2 (O+, 3 units, expiry 17-Nov-2026) under Near-Expiry Donations. O+ available stock is 4: prior 1 + new accepted 3; the expired 2 are excluded.

Blood requests: Request submission, review, approval and rejection were demonstrated. The latest invalid-request screenshot shows validation messages for required fields and no successful submission.

Blood issuance: A successful issue workflow was previously evidenced, including allocation and stock deduction. The attempt to access the issuance URL for rejected request TD5JKR1N93 displayed the rejected request detail page. This does not establish that the issue action is blocked; the case remains PARTIAL.

Access control: The incognito request to /reports/ redirected to /accounts/login/?next=/reports/, demonstrating unauthenticated route protection for that endpoint. Other authenticated hospital-role restrictions are shown for selected routes only.

Password storage: The terminal output for hospital_test showed a PBKDF2-SHA256 encoded hash and Is hashed: True. The password itself was not displayed.

Reports: Reports, history, filters and report totals were demonstrated in the baseline evidence set.

8. Findings, risks and follow-up actions

ID	Finding	Priority	Recommended action
F-01	Role/profile and Django Group authorization are not fully synchronized in the tested account setup; donor operations initially returned 403 until the required staff group was assigned.	High	Confirm intended role provisioning and ensure profile roles and group membership are consistently managed.
F-02	Rejected-request issuance protection is unresolved. The tested issuance URL displayed the rejected request detail page; no successful issue was attempted and no explicit denial was evidenced.	High	Have the application owner verify the route/view behaviour and perform a safe isolated test proving that rejected requests cannot be issued.
F-03	Failed issuance transaction rollback and insufficient-stock issuance failure are not demonstrated.	High	Use an isolated test database; verify request status, issue record, allocation and inventory remain consistent after failure.
F-04	Performance, concurrency, reliability, usability and portability targets lack fresh measurements.	Medium	Run repeatable timed, concurrent, recovery, usability and cross-browser tests if required by the assessment.
F-05	SQL injection, XSS, HTTPS transport and remaining security checks are not evidenced.	High	Perform authorized security testing in a controlled environment and retain tool output and test records.
F-06	The test evidence spans multiple runs and database states. Earlier report context referenced PostgreSQL while the latest local shell evidence used SQLite.	Medium	Keep run identifiers, database backend, timestamps and test data documented; do not compare totals across separate states.

9. STP exit-criteria assessment

STP exit criterion	Assessment	Basis
100% planned test cases executed	NOT MET	10 of 28 cases remain NOT RUN; 2 are PARTIAL.
At least 95% of executed tests passed	NOT DEMONSTRATED	No formal execution log and defensible executed-test denominator supplied; no pass rate calculated.
Zero open Critical / High defects	NOT DEMONSTRATED	High-priority authorization, rejected-request issuance and security follow-ups remain unresolved.
100% functional requirements traced to passed tests	NOT DEMONSTRATED	A complete requirement-level RTM with passed-test coverage has not been re-executed.
All safety/security tests passed	NOT MET / NOT DEMONSTRATED	Several security cases remain PARTIAL or NOT RUN.
Performance targets met	NOT DEMONSTRATED	No repeatable response-time or load measurements supplied.

10. Conclusion

The evidence demonstrates substantial core BBMS functionality, including donor and donation management, inventory calculations, expiry monitoring, blood request review, successful issuance, reports and selected access-control behaviour. The latest evidence also demonstrates invalid donor and request form validation, expired-stock exclusion, near-expiry donation display, unauthenticated report-route protection and password hashing.

However, the report does not claim complete system acceptance. Ten of the 28 STP cases are NOT RUN and two are PARTIAL. Rejected-request issuance protection remains unresolved, and performance, reliability, portability and several security objectives lack execution evidence. Therefore, the appropriate overall conclusion is PARTIAL SYSTEM VALIDATION; the STP exit criteria are not demonstrated.

Appendix A – Latest screenshot archive inventory

The latest archive current(1).zip contains 75 image files. The names below are reproduced from the archive.

- SYS-01_Login_Page.png
- SYS-03_Successful_Login.png
- SYS-05_Donor_Registration_403.png
- SYS-06_Donor_Registration_Form_After_Role_Assignment.png
- SYS-07_Donor_Registration_Success.png
- SYS-08_Donor_Database_Verification.png
- SYS-09_Donor_Search_Success.png
- SYS-10_Donor_History_Empty.png
- SYS-11_Donor_Update_Form.png
- SYS-12_Donor_Update_Success.png
- SYS-13_Accepted_Donation_Success.png
- SYS-14_Donation_Database_Verification.png
- SYS-15_Inventory_After_Accepted_Donation.png
- SYS-16_Inventory_Calculation_Verification.png
- SYS-17_Pending_Donation_Recorded.png
- SYS-18_Pending_Donation_Inventory_Exclusion.png
- SYS-19_Rejected_Donation_Recorded.png
- SYS-20_Donation_Status_Verification.png
- SYS-21_Rejected_Donation_Inventory_Exclusion.png
- SYS-22_Donor_Donation_History_All_Statuses.png
- SYS-23_Hospital_User_Login_Success.png
- SYS-24_Blood_Request_Form.png
- SYS-25_Blood_Request_Submitted.png
- SYS-26_Blood_Request_Database_Verification.png
- SYS-27_Staff_All_Blood_Requests.png
- SYS-28_Blood_Request_Review_Details.png
- SYS-29_Blood_Request_Approval_Success.png
- SYS-30_Blood_Request_Approval_Database_Verification.png
- SYS-31_Inventory_Unchanged_After_Approval.png
- SYS-32_Issue_Page_404_URL_Not_Registered.png
- SYS-33_Blood_Issuance_Page_Ready.png
- SYS-34_Blood_Issuance_Success.png
- SYS-35_Blood_Issuance_Database_Verification.png
- SYS-36_Inventory_After_Blood_Issuance.png
- SYS-37_Hospital_Request_Fulfilled_Status.png
- SYS-38_Rejection_Test_Request_Submitted.png

- SYS-39_Rejection_Request_Database_Verification.png
- SYS-40_Rejection_Request_Review_Page.png
- SYS-41_Blood_Request_Rejection_Success.png
- SYS-42_Blood_Request_Rejection_Database_Verification.png
- SYS-43_Inventory_Unchanged_After_Rejection.png
- SYS-44_Blood_Search_Page.png
- SYS-45_Blood_Search_O_Positive_Available.png
- SYS-46_Blood_Search_A_Positive_Unavailable.png
- SYS-47_Inventory_Dashboard_Final_State..png
- SYS-48_Reports_Dashboard_Initial_View..png
- SYS-49_Reports_Donation_Request_and_Issue_History.png
- SYS-50_Reports_Database_Cross_Verification.png
- SYS-51_Reports_Filter_Control.png
- SYS-52_Reports_Single_Day_Date_Filter.png
- SYS-53_Reports_O_Positive_Filter.png
- SYS-54_Reports_Pending_Donation_Filter.png
- SYS-55_Reports_Pending_Donation_History_Filtered.png
- SYS-56_Reports_Rejected_Request_Filter.png
- SYS-57_Reports_Rejected_Request_History_Filtered.png
- SYS-58_Reports_Reset_Functionality.png
- SYS-59_Hospital_Access_Denied_Pending_Requests.png
- SYS-60_Hospital_Access_Denied_All_Requests.png
- SYS-61_Hospital_Access_Denied_Blood_Issuance.png
- SYS-62_Hospital_Access_Denied_Donor_Registration.png
- SYS-63_Hospital_Access_Denied_Donor_Search.png
- SYS-64_Hospital_Inventory_Access_Allowed.png
- SYS-65_Hospital_Access_Denied_Reports.png
- SYS-66_Hospital_Access_Denied_Django_Admin.png
- SYS-67_Admin_Login_Success.png
- SYS-68_Admin_Reports_Access_Allowed.png
- SYS-69_Admin_Donor_Search_Access_Allowed.png
- SYS-NEW-01_Invalid_Request.png
- SYS-NEW-02_Expired_Donation_Registration.png
- SYS-NEW-02_Expired_Donation_Registration2.png
- SYS-NEW-03_Expired_Stock_Exclusion.png
- SYS-NEW-04_Unauthenticated_Access.png
- SYS-NEW-05_Password_Hash_Verification.png
- SYS-NEW-06_Near_Expiry_Donation_Registration.png
- SYS-NEW-06_Near_Expiry_Stock_Verification.png

Appendix B – Additional individually shared evidence

- Donor required-field validation screenshot (individual upload in conversation; not part of the numbered archive).
- Donor invalid-phone validation screenshot (individual upload in conversation; not part of the numbered archive).
- Rejected-request details / issuance URL attempt for request TD5JKR1N93 (individual upload in conversation).
- Terminal output showing PBKDF2-SHA256 password hash and Is hashed: True (text output in conversation).

Appendix C – Historical automated results

The previous v1.4 report recorded 131 Django tests, 129 passed, 2 skipped and 89% code coverage. The skipped areas included browser portability and concurrent blood issuance requiring PostgreSQL. These values are historical and are not represented as freshly rerun results in this v3.1 assessment.

Appendix D – Evidence handling

The latest screenshot archive should be retained alongside this report. The report references screenshots by filename rather than embedding all 75 images, to keep the document readable and manageable. The original archive is the supporting evidence package. Do not remove or alter the original evidence files after submission.